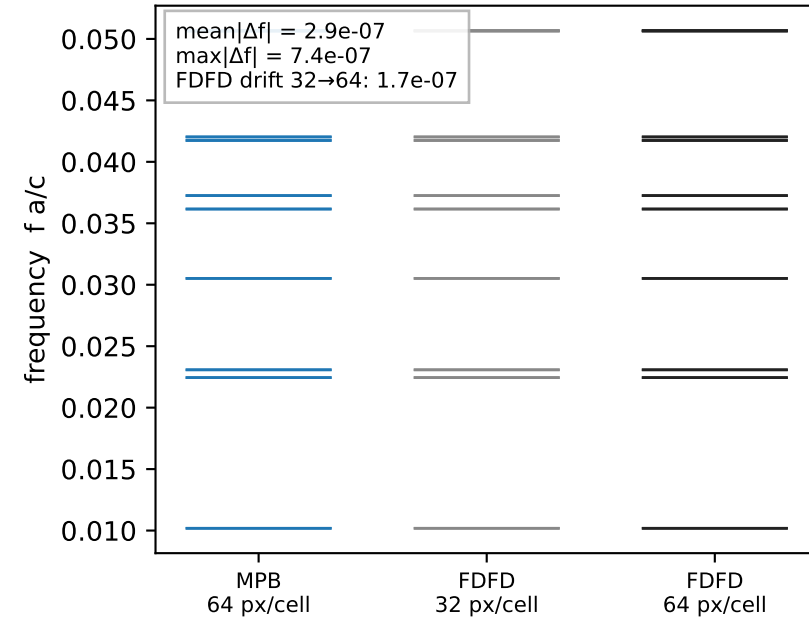
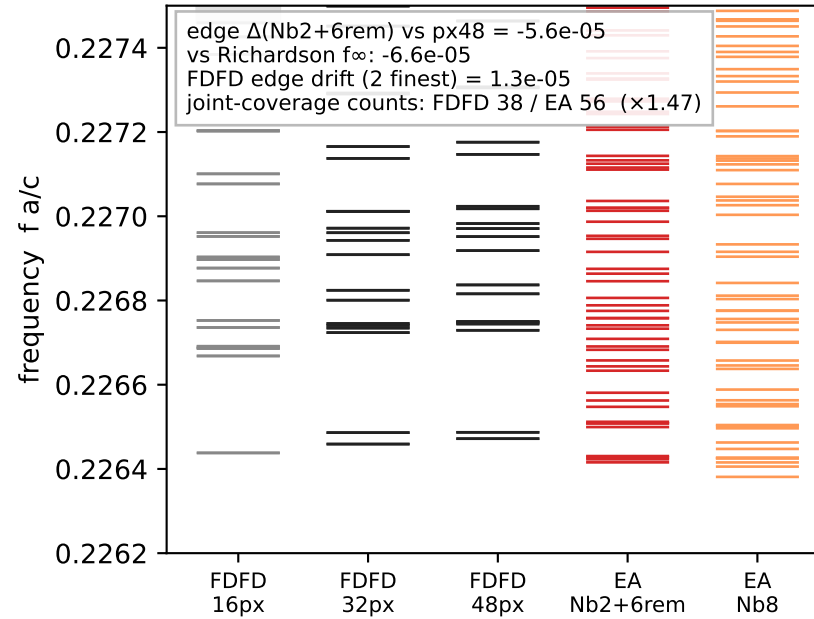


Strict MPB / FDFD / EA eigenvalue comparison — square CML, TM at X (triangle protocol)

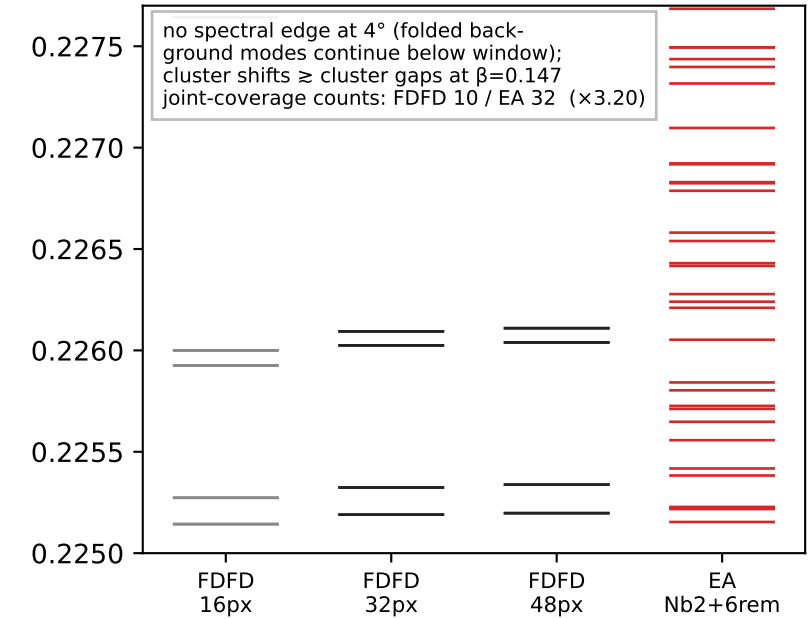
A MPB ↔ FDFD, exact methods
(29,1) $\theta=3.950^\circ$, supercell $k=(\frac{1}{2},0)$, lowest 20



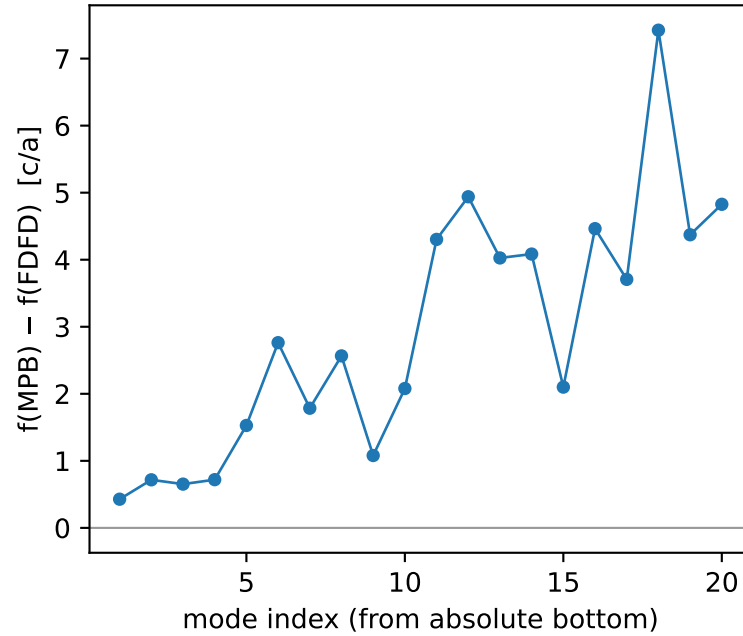
B FDFD ↔ EA, band edge, IN-ZONE
(57,1) $\theta=2.010^\circ$, $\beta=0.075$, $k=Q_X$



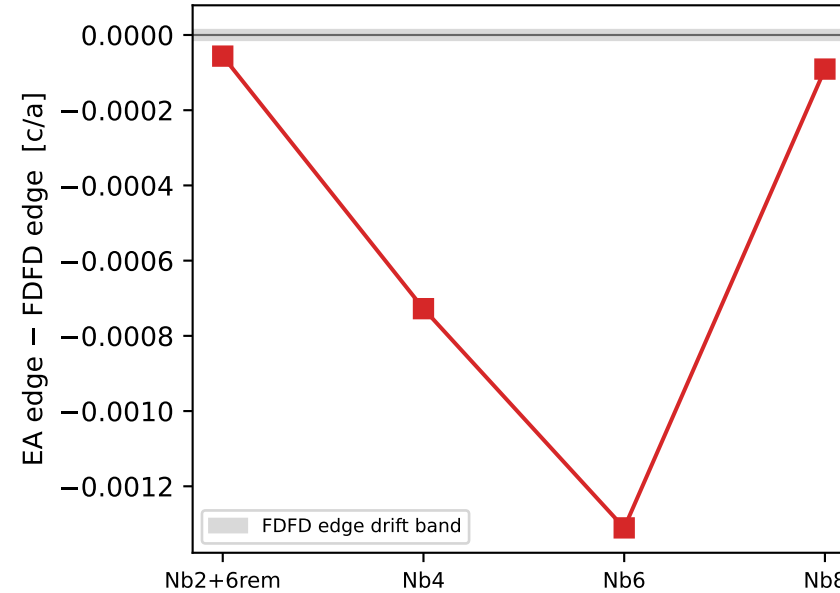
C FDFD ↔ EA, band edge, MARGINAL
(29,1) $\theta=3.950^\circ$, $\beta=0.147$, $k=Q_X$



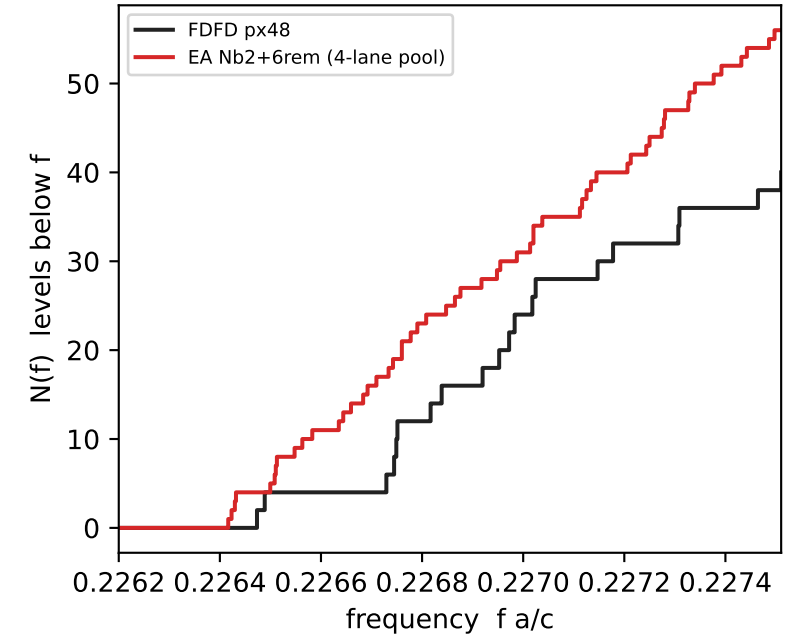
A' per-mode residual, index-aligned



D Nb ladder at 2°: edge error vs retained bands
(non-monotonic — η^2 -truncated operator is non-variational)



E level-counting functions, 2° edge
EA ladder squeezed: levels inside FDFD gaps



Square lattice (const a), TM (E_z) | rods $r/a=0.2$, $\epsilon=8.9$ in $\epsilon_{bg}=1.0$ | twisted bilayer, both layers in-plane
(57,1): $\theta=2.0102^\circ$, $L=57.009a$, 3250 cells | (29,1): $\theta=3.9499^\circ$, $L=29.017a$, 842 cells | accuracy zone $\beta=\theta_{rad}/y$, $\gamma(X,0\rightarrow1)=0.468$: $2^\circ-\beta=0.075$ (in-zone), $4^\circ-\beta=0.147$ (marginal)
EA: blaze2d V4 exact-TM (Berry-free), $k_0 = \text{monolayer } X = (\frac{1}{2},0) \cdot 2\pi/a$, band 0, $\lambda_{ref}=2.35584$ ($f_{ref}=0.24428$) | registry 128^2 , envelope $N_s=128^2$, fd_order 4 | shift-invert $f_0=0.2270$, 15 modes/lane
EA k -lanes: $\Gamma_m, X1_m, X2_m, M_m$ (2×2 CML fold), pooled, no valley doubling | FDFD: $L=\epsilon^{-1/2}(\sum g^{ab}D^{\dagger}D)\epsilon^{-1/2}$, subpixel ϵ -avg $N_{sub}=8$, CHOLMOD shift-invert, tol $1e-10$
FDFD bottom runs: $Q_X=(\pi,0) \equiv \text{supercell } k=(\frac{1}{2},\frac{1}{2})$, $\sigma_\omega=0.2270$ | MPB: res 64/cell, 20 bands, run_tm, supercell $k=(\frac{1}{2},0)$ — FDFD leg A matches MPB's k exactly
STRICT protocol: sorted ladders, no Hungarian/assignment matching; panel A index-aligned from absolute mode 1
($m,1$) supercell = $2 \times \text{CENTERED cell}$: $\tau=(L1+L2)/2 \in \text{both layers' lattices (verified exactly)} \rightarrow \text{FDFD } \times 2 \text{ pairs} = \text{primitive-cell folding} = X \otimes X'$ valleys at Q_X ; EA lanes $\{\Gamma_m, M_m\} = \text{valley } X$, $\{X1_m, X2_m\} = \text{valley } X'$ (C4-paired, verified to $2e-5$)